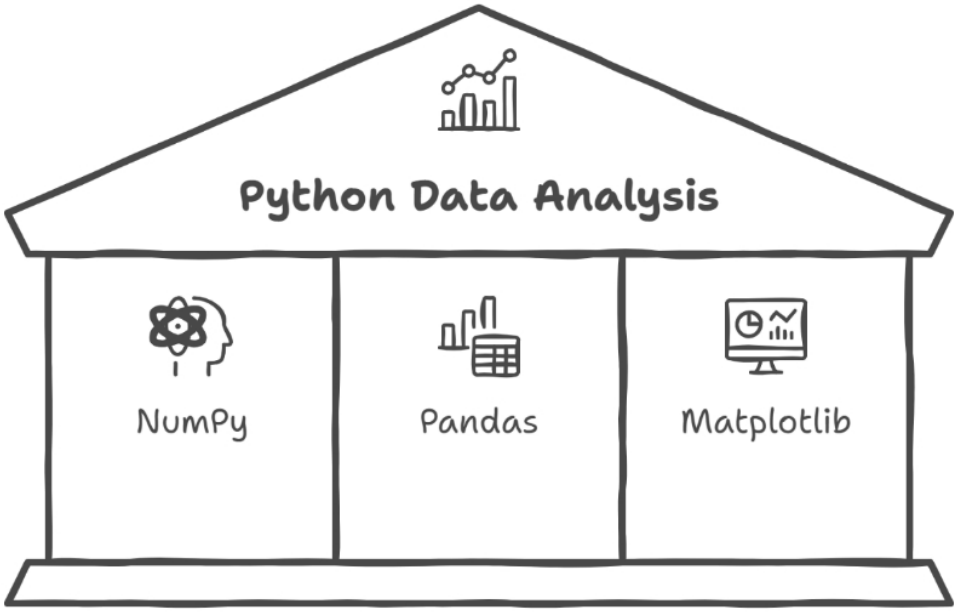


1. NumPy

NumPy forms the foundation of numerical computing in Python. It provides high-performance tools for working with large, multidimensional arrays and matrices. NumPy significantly outperforms Python's native lists in terms of speed and efficiency, especially when dealing with large datasets. Its extensive library of mathematical functions allows developers to perform computations such as linear algebra, Fourier transforms, and statistical operations with minimal code. NumPy is widely used as the backbone of other Python libraries like Pandas, Matplotlib, and Scikit-learn, making it an essential skill for data scientists and ML practitioners.

Example:

```
import numpy as np
array = np.array([10, 20, 30, 40])
print("Array:", array)
print("Mean:", np.mean(array))
print("Standard Deviation:", np.std(array)) print("Sum:", np.sum(array))
print("Squared Values:", np.square(array)) Features:
```

- High-speed operations on arrays and matrices.